

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250278124

Kind Code

A1

Publication Date

September 04, 2025

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FOLDABLE ELECTRONIC DEVICE INCLUDING HINGE LOCKING STRUCTURE

Abstract

An example electronic device may include a first housing including a first surface, a second housing including a second surface, a third housing including a third surface, a first hinge structure rotatably connecting the second housing and the first housing, a second hinge structure rotatably connecting the first housing and the third housing, a flexible display disposed on the first surface, the second surface and the third surface, a first magnet disposed in the second housing, and a hinge locking structure which, in an abnormal state of the electronic device in which a portion of the first surface is adjacent to the second surface, faces the first magnet and which includes a pin insertable into a groove formed in at least one shaft in the second hinge structure.

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Family ID: 89323913

Appl. No.: 19/205459

Filed: May 12, 2025

Foreign Application Priority Data

KR 10-2022-0078823

Jun. 28, 2022

KR 10-2022-0098134

Aug. 05, 2022

Related U.S. Application Data

parent US continuation 18190632 20230327 parent-grant-document US 12298819 child US 19205459

parent WO continuation PCT/KR2023/003718 20230321 PENDING child US 18190632

Publication Classification

Int. Cl.: G06F1/16 (20060101)

U.S. Cl.:

CPC G06F1/1681 (20130101); G06F1/1616 (20130101); G06F1/1652 (20130101);
G06F1/1677 (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 18/190,632, filed Mar. 27, 2023, which is a continuation of International Application No. PCT/KR2023/003718, designating the United States, filed on Mar. 21, 2023, in the Korean Intellectual Property Receiving Office and claiming priority to Korean Patent Application No. 10-2022-0078823, filed on Jun. 28, 2022 in the Korean Intellectual Property Office and to Korean Patent Application No. 10-2022-0098134, filed on Aug. 5, 2022 in the Korean Intellectual Property Office. The disclosures of each of these applications are incorporated by reference herein in their entireties.

BACKGROUND

Field

[0002] The disclosure relates to a foldable electronic device including a hinge locking structure.

Description of Related Art

[0003] An electronic device including a large screen display may increase user utilization. As demand for electronic devices having high portability increases, the electronic devices may be provided with a deformable display. The deformable display may be deformable in a slidable manner, foldable manner, or rollable manner.

SUMMARY

[0004] A deformable display may have structurally weak rigidity. The deformable display may be damaged during use or by use contrary to an intended usage. An electronic device described herein may provide a structure and a method for preventing damage to the deformable display.

[0005] The technical problems to be achieved in this disclosure are not limited to those described above, and other technical problems not mentioned herein will be clearly understood from the following description.

[0006] According to an example embodiment, an electronic device may include a first housing including a first surface, a second housing including a second surface, and a third housing including a third surface. The electronic device may further include a first hinge structure (including, e.g., a hinge) rotatably connecting the first housing and a side surface of the second housing facing a first side surface of the first housing. The electronic device may further include a second hinge structure (including, e.g., a hinge) rotatably connecting the first housing and a side surface of the third housing facing a second side surface of the first housing opposite to the first side surface. The electronic device may further include a flexible display disposed on the first surface, the second surface and the third surface. The electronic device may include a first magnet disposed in the second housing, and a hinge locking structure (including, e.g., a hinge lock) which, in an abnormal state of the electronic device in which a portion of the first surface is adjacent to the second surface, faces the first magnet and which includes a pin insertable into a groove formed in at least one shaft in the second hinge structure. The first hinge structure and the second hinge structure may be configured to provide an unfolded state in which the first surface, the second

surface, and the third surface face the same direction or a folded state in which at least one of the first surface, the second surface, or the third surface faces different directions. The hinge locking structure may be configured such that the pin is inserted into the groove by a force of the first magnet when the electronic device is in the abnormal state, and the pin is disengaged from the groove as the electronic device changes from the abnormal state to the unfolded state.

[0007] According to an embodiment, an electronic device may include a first housing including a first surface, a second housing including a second surface, and a third housing including a third surface. The electronic device may further include a first hinge structure (including, e.g., a hinge) rotatably connecting the first housing and a side surface of the second housing facing a first side surface of the first housing. The electronic device may further include a second hinge structure (including, e.g., a hinge) rotatably connecting the first housing and a side surface of the third housing facing a second side surface of the first housing opposite to the first side surface. The electronic device may further include a flexible display disposed on the first surface, the second surface and the third surface. The electronic device may further include a first sensor disposed in the second housing and configured to obtain first data related to a posture of the second housing, and a second sensor disposed in the third housing and configured to obtain second data related to a posture of the third housing. The electronic device may include a first magnet disposed in the second housing, and a hinge locking structure (including, e.g., a hinge lock) which, in an abnormal state of the electronic device in which a portion of the first surface is adjacent to the second surface, faces the first magnet and includes a pin insertable into a groove formed in at least one shaft in the second hinge structure. The first hinge structure and the second hinge structure may be configured to provide an unfolded state in which the first surface, the second surface, and the third surface face the same direction or a folded state in which at least one of the first surface, the second surface, or the third surface face different directions. An overlapped area of the flexible display and the first hinge structure may be wider than an overlapped area of the flexible display and the second hinge structure, in the unfolded state when the flexible display is viewed from above.

[0008] According to an embodiment, a foldable electronic device including a hinge locking structure can reduce damage to a display or the electronic device due to folding that is contrary to intended usage.

[0009] The effects that can be obtained from the present disclosure are not limited to those described above, and any other effects not mentioned herein will be clearly understood from the following description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] The above and/or other aspects of the disclosure will be more apparent by describing certain embodiments of the disclosure with reference to the accompanying drawings, in which:

[0011] FIG. 1 is a block diagram of an example electronic device in a network environment according to various embodiments;

[0012] FIG. 2A illustrates an example of an unfolded state of an example electronic device according to various embodiments;

[0013] FIG. 2B illustrates an example of a fully folded state of an example electronic device according to various embodiments;

[0014] FIG. 3A illustrates an example of arrangement of a hinge locking structure in an unfolded state of an example electronic device according to various embodiments;

[0015] FIG. 3B illustrates an example hinge locking structure in an unfolded state of an example electronic device according to various embodiments;

[0016] FIG. 4A illustrates an example of arrangement of a hinge locking structure in a third state of

an example electronic device according to various embodiments;

[0017] FIG. 4B illustrates an example hinge locking structure in a third state of an example electronic device according to various embodiments;

[0018] FIG. 5A illustrates an example of arrangement of a hinge locking structure when a third housing moves in a fourth state of an example electronic device according to various embodiments;

[0019] FIG. 5B illustrates an example hinge locking structure when a third housing moves in a fourth state of an example electronic device according to various embodiments;

[0020] FIG. 6 is a block diagram of an example electronic device according to various embodiments; and

[0021] FIG. 7 is a flowchart illustrating an example operation of detecting an abnormal state of an example electronic device according to various embodiments.

DETAILED DESCRIPTION

[0022] FIG. 1 is a block diagram illustrating an example electronic device **101** in a network environment **100** according to various embodiments.

[0023] Referring to FIG. 1, the electronic device **101** in the network environment **100** may communicate with an electronic device **102** via a first network **198** (e.g., a short-range wireless communication network), or at least one of an electronic device **104** or a server **108** via a second network **199** (e.g., a long-range wireless communication network). According to an embodiment, the electronic device **101** may communicate with the electronic device **104** via the server **108**.

According to an embodiment, the electronic device **101** may include a processor **120**, memory **130**, an input module **150**, a sound output module **155**, a display module **160**, an audio module **170**, a sensor module **176**, an interface **177**, a connecting terminal **178**, a haptic module **179**, a camera module **180**, a power management module **188**, a battery **189**, a communication module **190**, a subscriber identification module (SIM) **196**, or an antenna module **197**. In various embodiments, at least one of the components (e.g., the connecting terminal **178**) may be omitted from the electronic device **101**, or one or more other components may be added in the electronic device **101**. In some embodiments, some of the components (e.g., the sensor module **176**, the camera module **180**, or the antenna module **197**) may be implemented as a single component (e.g., the display module **160**).

[0024] The processor **120** may execute, for example, software (e.g., a program **140**) to control at least one other component (e.g., a hardware or software component) of the electronic device **101** coupled with the processor **120**, and may perform various data processing or computation.

According to an embodiment, as at least part of the data processing or computation, the processor **120** may store a command or data received from another component (e.g., the sensor module **176** or the communication module **190**) in volatile memory **132**, process the command or the data stored in the volatile memory **132**, and store resulting data in non-volatile memory **134**. According to an embodiment, the processor **120** may include a main processor **121** (e.g., a central processing unit (CPU) or an application processor (AP)), or an auxiliary processor **123** (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **121**. For example, when the electronic device **101** includes the main processor **121** and the auxiliary processor **123**, the auxiliary processor **123** may be adapted to consume less power than the main processor **121**, or to be specific to a specified function. The auxiliary processor **123** may be implemented as separate from, or as part of the main processor **121**.

[0025] The auxiliary processor **123** may control at least some of functions or states related to at least one component (e.g., the display module **160**, the sensor module **176**, or the communication module **190**) among the components of the electronic device **101**, instead of the main processor **121** while the main processor **121** is in an inactive (e.g., sleep) state, or together with the main processor **121** while the main processor **121** is in an active state (e.g., executing an application). According to an embodiment, the auxiliary processor **123** (e.g., an image signal processor or a

communication processor) may be implemented as part of another component (e.g., the camera module **180** or the communication module **190**) functionally related to the auxiliary processor **123**. According to an embodiment, the auxiliary processor **123** (e.g., the neural processing unit) may include a hardware structure specified for artificial intelligence model processing. An artificial intelligence model may be generated by machine learning. Such learning may be performed, e.g., by the electronic device **101** where the artificial intelligence is performed or via a separate server (e.g., the server **108**). Learning algorithms may include, but are not limited to, e.g., supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural network may be a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted Boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent deep neural network (BRDNN), deep Q-network or a combination of two or more thereof but is not limited thereto. The artificial intelligence model may, additionally or alternatively, include a software structure other than the hardware structure.

[0026] The memory **130** may store various data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**.

[0027] The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.

[0028] The input module **150** may receive a command or data to be used by another component (e.g., the processor **120**) of the electronic device **101**, from the outside (e.g., a user) of the electronic device **101**. The input module **150** may include, for example, a microphone, a mouse, a keyboard, a key (e.g., a button), or a digital pen (e.g., a stylus pen).

[0029] The sound output module **155** may output sound signals to the outside of the electronic device **101**. The sound output module **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record. The receiver may be used for receiving incoming calls. According to an embodiment, the receiver may be implemented as separate from, or as part of the speaker.

[0030] The display module **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display module **160** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to an embodiment, the display module **160** may include a touch sensor adapted to detect a touch, or a pressure sensor adapted to measure the intensity of force incurred by the touch.

[0031] The audio module **170** may convert a sound into an electrical signal and vice versa. According to an embodiment, the audio module **170** may obtain the sound via the input module **150**, or output the sound via the sound output module **155** or a headphone of an external electronic device (e.g., an electronic device **102**) directly (e.g., wiredly) or wirelessly coupled with the electronic device **101**.

[0032] The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal or data value corresponding to the detected state. According to an embodiment, the sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

[0033] The interface **177** may support one or more specified protocols to be used for the electronic device **101** to be coupled with the external electronic device (e.g., the electronic device **102**) directly (e.g., wiredly) or wirelessly. According to an embodiment, the interface **177** may include,

for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

[0034] A connecting terminal **178** may include a connector via which the electronic device **101** may be physically connected with the external electronic device (e.g., the electronic device **102**). According to an embodiment, the connecting terminal **178** may include, for example, a HDMI connector, a USB connector, a SD card connector, or an audio connector (e.g., a headphone connector).

[0035] The haptic module **179** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or a movement) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an embodiment, the haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

[0036] The camera module **180** may capture a still image or moving images. According to an embodiment, the camera module **180** may include one or more lenses, image sensors, image signal processors, or flashes.

[0037] The power management module **188** may manage power supplied to the electronic device **101**. According to an embodiment, the power management module **188** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

[0038] The battery **189** may supply power to at least one component of the electronic device **101**. According to an embodiment, the battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

[0039] The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101** and the external electronic device (e.g., the electronic device **102**, the electronic device **104**, or the server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more communication processors that are operable independently from the processor **120** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment, the communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device via the first network **198** (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or the second network **199** (e.g., a long-range communication network, such as a legacy cellular network, a 5G network, a next-generation communication network, the Internet, or a computer network (e.g., LAN or wide area network (WAN))). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **192** may identify and authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **196**.

[0040] The wireless communication module **192** may support a 5G network, after a 4G network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module **192** may support a high-frequency band (e.g., the mmWave band) to achieve, e.g., a high data transmission rate. The wireless communication module **192** may support various technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (massive MIMO), full dimensional

MIMO (FD-MIMO), array antenna, analog beam-forming, or large scale antenna. The wireless communication module **192** may support various requirements specified in the electronic device **101**, an external electronic device (e.g., the electronic device **104**), or a network system (e.g., the second network **199**). According to an embodiment, the wireless communication module **192** may support a peak data rate (e.g., 20 Gbps or more) for implementing eMBB, loss coverage (e.g., 164 dB or less) for implementing mMTC, or U-plane latency (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL), or a round trip of 1 ms or less) for implementing URLLC.

[0041] The antenna module **197** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device) of the electronic device **101**. According to an embodiment, the antenna module **197** may include an antenna including a radiating element composed of or including a conductive material or a conductive pattern formed in or on a substrate (e.g., a printed circuit board (PCB)). According to an embodiment, the antenna module **197** may include a plurality of antennas (e.g., array antennas). In such a case, at least one antenna appropriate for a communication scheme used in the communication network, such as the first network **198** or the second network **199**, may be selected, for example, by the communication module **190** (e.g., the wireless communication module **192**) from the plurality of antennas. The signal or the power may then be transmitted or received between the communication module **190** and the external electronic device via the selected at least one antenna. According to an embodiment, another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as part of the antenna module **197**.

[0042] According to various embodiments, the antenna module **197** may form a mmWave antenna module. According to an embodiment, the mmWave antenna module may include a printed circuit board, a RFIC disposed on a first surface (e.g., the bottom surface) of the printed circuit board, or adjacent to the first surface and capable of supporting a designated high-frequency band (e.g., the mmWave band), and a plurality of antennas (e.g., array antennas) disposed on a second surface (e.g., the top or a side surface) of the printed circuit board, or adjacent to the second surface and capable of transmitting or receiving signals of the designated high-frequency band.

[0043] At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

[0044] According to an embodiment, commands or data may be transmitted or received between the electronic device **101** and the external electronic device **104** via the server **108** coupled with the second network **199**. Each of the electronic devices **102** or **104** may be a device of a same type as, or a different type, from the electronic device **101**. According to an embodiment, all or some of operations to be executed at the electronic device **101** may be executed at one or more of the external electronic devices **102**, **104**, or **108**. For example, if the electronic device **101** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device **101**. The electronic device **101** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device **101** may provide ultra low-latency services using, e.g., distributed computing or mobile edge computing. In an embodiment, the external electronic device **104** may include an internet-of-things (IoT) device. The server **108** may be an intelligent server using machine learning and/or a neural network. According to an embodiment, the external electronic device **104** or the

server **108** may be included in the second network **199**. The electronic device **101** may be applied to intelligent services (e.g., smart home, smart city, smart car, or healthcare) based on 5G communication technology or IoT-related technology.

[0045] FIG. 2A illustrates an example of an unfolded state of an example electronic device according to various embodiments. FIG. 2B illustrates an example of a fully folded state of an example electronic device according to various embodiments.

[0046] A display (e.g., the display module **160** of FIG. 1) of an electronic device (e.g., the electronic device **101** of FIG. 1) may be folded at least once. The display may be referred to, for example, as a foldable display or a flexible display in terms of having a foldable structure. The electronic device may be referred to, for example, as a foldable electronic device or a flexible electronic device in terms of including a foldable display.

[0047] Referring to FIG. 2A, an electronic device **200** may be unfolded in a first state **200a**. The electronic device **200** may include a first housing **210**, a second housing **220**, and a third housing **230**. The first housing **210** may include a first surface **210a**. The second housing **220** may include a second surface **220a**. The third housing **230** may include a third surface **230a**. The first state **200a** may be a state in which the first surface **210a**, the second surface **220a**, and the third surface **230a** face substantially the same first direction (e.g., the z-axis direction). The first state **200a** may be referred to, for example, as an unfolded state in which the first housing **210**, the second housing **220**, and the third housing **230** are unfolded. In the first state **200a**, the first housing **210**, the second housing **220**, and the third housing **230** may be substantially planar. For example, since the first housing **210**, the second housing **220**, and the third housing **230** include a substantially planar surface, the first surface **210a**, the second surface **220a**, and the third surface **230a** may be planar.

[0048] The electronic device **200** may include a first hinge structure **202** (**202a** and **202b**) and a second hinge structure **203** (**203a** and **203b**). The first hinge structure **202** may include a plurality of first hinge structures **202a** and **202b**. The second hinge structure **203** may include a plurality of second hinge structures **203a** and **203b**. The first hinge structures **202a** and **202b** may be disposed between the first housing **210** and the second housing **220**. The first hinge structures **202a** and **202b** may rotatably connect the second housing **220** to the first housing **210**. For example, the second housing **220** may rotate with respect to the first housing **210** based on a first folding shaft **f1**, through the first hinge structures **202a** and **202b**. A first angle θ_1 between the first housing **210** and the second housing **220** may be approximately 0 degrees to 180 degrees, through the rotation of the second housing **220**. The second hinge structures **203a** and **203b** may be disposed between the first housing **210** and the third housing **230**. The second hinge structures **203a** and **203b** may rotatably connect the third housing **230** to the first housing **210**. For example, the third housing **230** may rotate with respect to the first housing **210** based on a second folding shaft **f2**, through the second hinge structures **203a** and **203b**. The second angle θ_2 between the first housing **210** and the third housing **230** may be approximately 0 degrees to 180 degrees, through the rotation of the third housing **230**.

[0049] The first hinge structures **202a** and **202b** may be surrounded by the first hinge cover **250**. One (e.g., **202a**) of the first hinge structures **202a** and **202b** may be disposed at one end of the first hinge cover **250**, and another (e.g., **202b**) may be disposed at another end of the first hinge cover **250**. The second hinge structures **203a** and **203b** may be surrounded by a second hinge cover **260**. One (e.g., **203a**) of the second hinge structures **203a** and **203b** may be disposed at one end of the second hinge cover **260**, and another (e.g., **203b**) may be disposed at another end of the second hinge cover **260**.

[0050] The electronic device **200** may further include a flexible display **240**. The flexible display **240** may be disposed on one surface formed by the first housing **210**, the second housing **220**, and the third housing **230**. The flexible display **240** may be disposed on the first surface **210a**, the second surface **220a**, and the third surface **230a**. The flexible display **240** may include a first display area **240a**, a second display area **240b**, and a third display area **240c**. The first display area

240a may be disposed on the first surface **210a** of the first housing **210**. The first display area **240a** may be an area coupled to the first surface **210a** and not deformed. The second display area **240b** may be disposed on the second surface **220a** of the second housing **220**. The second display area **240b** may be an area coupled to the second surface **220a** and not deformed. The third display area **240c** may be disposed on the third surface **230a** of the third housing **230**. The third display area **240c** may be an area coupled to the third surface **230a** and not deformed. The first display area **240a**, the second display area **240b**, and the third display area **240c** may be referred to, for example, as planar areas, in terms of forming substantially planar surface.

[0051] The flexible display **240** may further include a fourth display area **240d** and a fifth display area **240e**. The fourth display area **240d** may be disposed on the first hinge cover **250**. The fourth display area **240d** may be coupled to the first hinge cover **250** or the first hinge structures **202a** and **202b**. For example, the fourth display area **240d** may be supported by a hinge plate (not illustrated) included in the first hinge structures **202a** and **202b**. The fourth display area **240d** may be deformed as the second housing **220** rotates based on the first folding shaft **f1** with respect to the first housing **210**.

[0052] The fifth display area **240e** may be disposed on the second hinge cover **260**. The fifth display area **240e** may be coupled to the second hinge cover **260** or the second hinge structures **203a** and **203b**. For example, the fifth display area **240e** may be supported by a hinge plate (not illustrated) included in the second hinge structures **203a** and **203b**. The fifth display area **240e** may be deformed as the third housing **230** rotates based on the second folding shaft **f2** with respect to the first housing **210**.

[0053] The fourth display area **240d** may be larger than the fifth display area **240e**. For example, a first distance **d1**, which is the width of the fourth display area **240d**, may be longer (greater) than a second distance **d2**, which is the width of the fifth display area **240e**. When the electronic device **200** is converted into the folded state, the radius of curvature of the fourth display area **240d** may be greater than the radius of curvature of the fifth display area **240e**.

[0054] The electronic device **200** may further include first magnets **201a** and **201b**. The first magnets **210a** and **210b** may be disposed in the second housing **220**. For example, the first magnets **210a** may be disposed on the side surface of the second housing **220**. The first magnets **210a** may be disposed closer to another side surface facing the one side surface than the one side surface of the second housing **220** facing the first hinge cover **250**. The first magnets **210a** may at least partially overlap an inactive area disposed along the periphery of the second display area **240b** when the flexible display **240** is viewed from above. The first magnets **201a** and **201b** may be disposed to face the second hinge structures **203a** and **203b**, when the second housing **220** is folded by rotating along the first folding shaft **f1**.

[0055] Referring to FIG. 2B, the electronic device **200** may be folded in a second state **200b**. In the second state **200b**, at least one of the first surface **210a**, the second surface **220a**, or the third surface **230a** may face different directions. The third display area **240c** corresponding to the third surface **230a** may face substantially the same direction as the direction in which the second display area **240b** corresponding to the second surface **220a** is directed (e.g., the $-z$ axis direction). The third display area **240c** may face a direction distinct from a direction in which the first display area **240a** corresponding to the first surface **210a** is directed. For example, a direction in which the third display area **240c** is directed (e.g., the $-z$ axis direction) may be opposite to a direction in which the first display area **240a** is directed (e.g., the $+z$ axis direction). The third display area **240c** may face the first display area **240a**. Since the first housing **210**, the second housing **220**, and the third housing **230** are folded, the second state **200b** may be referred to, for example, as a folded state. In the second state **200b**, the first housing **210**, the second housing **220**, and the third housing **230** may face each other. For example, in the second state **200b**, the first housing **210**, the third housing **230**, and the second housing **220** may be stacked in the first direction (e.g., the z axis direction).

[0056] While changing from the first state **200a** to the second state **200b**, the second housing **220**

may rotate with respect to the first housing **210** based on the first folding shaft **f1**, through the first hinge structure **202**. In the second state **200b**, the second housing **220** may face the third housing **230**. For example, in the second state **200b**, the second display area **240b** disposed on the second housing **220** may face a rear surface **231** of the third housing **230**.

[0057] Through the second hinge structure **203**, the third housing **230** may rotate with respect to the first housing **210** based on the second folding shaft **f2**. In the second state **200b**, the third housing **230** may face the first housing **210**. For example, in the second state **200b**, the third display area **240c** disposed on the third housing **230** may face the first display area **240a** disposed on the first housing **210**. The side surface **232** of the third housing **230** may face the fourth display area **240d**.

[0058] In the second state **200b**, the overlapped area of the flexible display **240** and the first hinge structure **202** when the first hinge cover **250** is viewed from above may be wider than the overlapped area of the flexible display **240** and the second hinge structure **203** when the second hinge cover **260** is viewed from above. The first distance **d1**, which is the width of the first hinge cover **250** corresponding to the fourth display area **240d**, may be longer (greater) than the second distance **d2**, which is the width of the second hinge cover **260** corresponding to the fifth display area **240e** (e.g., the fifth display area **240e** of FIG. 2A). The difference between the first distance **d1** and the second distance **d2** may correspond to the thickness of the second housing **220**.

[0059] Based on the difference between the first distance **d1** and the second distance **d2**, a preferentially foldable housing may be determined among the second housing **220** and the third housing **230**. For example, the third housing **230** may preferentially rotate with respect to the first housing **210** based on the second folding shaft **f2**. After the third housing **230** is folded, the second housing **220** may rotate with respect to the first housing **210** based on the first folding shaft **f1**. Since the width of the second hinge structure **203** is narrower than the width of the first hinge structure **202**, the third housing **230** may be folded to face the first housing **210**. For example, since the width of the first hinge cover **250** is wider than that of the second hinge cover **260**, the third housing **230** may be folded to face the first housing **210** based on the operation of the second hinge structure **203** in the second hinge cover **260**. Since the width of the first hinge structure **202** is wider than the width of the second hinge structure **203**, the second housing **220** may be folded to face the rear surface **231** of the third housing **230**. For example, since the width of the first hinge cover **250** is wider than the width of the second hinge cover **260**, the second housing **220** may be folded to face the rear surface **231** of the third housing **230** based on the operation of the first hinge structure **202** in the first hinge cover **250**.

[0060] According to the above-described embodiment, the electronic device **200** may provide the flexible display **240** that may be folded at least once by including a plurality of hinge structures **202** and **203**. The flexible display **240** may be folded several times, through the different width of hinge structures **202** and **203**.

[0061] FIG. 3A illustrates an example of arrangement of a hinge locking structure in an unfolded state of an example electronic device according to various embodiments. FIG. 3B illustrates an example hinge locking structure in an unfolded state of an example electronic device according to various embodiments.

[0062] Referring to FIG. 3A, the electronic device **200** may be unfolded in the first state **200a**. The flexible display **240** may be disposed on a planar surface formed by the first housing **210**, the second housing **220**, and the third housing **230**.

[0063] The first hinge structure **202** may rotatably connect one side surface **221** of the second housing **220** facing the first side surface **211** of the first housing **210** to the first housing **210**. The first angle θ_1 between the first housing **210** and the second housing **220** may be approximately 0 degrees to 180 degrees, through the rotation of the second housing **220**. The second hinge structure **203** may rotatably connect one side surface **231** of the third housing **230** facing the second side surface **212** opposite to the first side surface **211** to the first housing **210**. The second angle θ_2

between the first housing **210** and the third housing **230** may be approximately 0 degree to 180 degrees, through rotation of the third housing **230**.

[0064] The first magnet **201** may be disposed in the second housing **220**. The first magnet **201** may be disposed on a periphery portion **229** of the second housing **220** including another side surface **222** of the second housing **220** facing one side surface **221** of the second housing **220**.

[0065] A hinge locking structure **330** may be disposed in the second hinge structure **203**. According to an embodiment, the second hinge structure **203** may include the hinge locking structure **330**. The second hinge structure **203** may further include a physical structure for the hinge locking structure **330**.

[0066] Referring to FIG. **3B**, the second hinge structure **203** may include at least one shaft **310** and a shaft holder **320**. For example, the second hinge structure **203** may rotate the third housing **230** with respect to the first housing **210** based on the second folding shaft **f2**.

[0067] The at least one shaft **310** may rotate through a change in a folded state or an unfolded state of the electronic device **200**. The shaft holder **320** may rotatably support at least one shaft **310**. The at least one shaft **310** may rotate a gear (not illustrated). The shafts in the second hinge structure **203** may be interlocked by the gear.

[0068] The shaft holder **320** may include a first holder **320a** supporting the side surface of the shaft **310** and a second holder **320b** supporting the lower part of the shaft **310**. The shaft holder **320** may include a space **321** for the hinge locking structure **330**. The space **321** may guide the movement of the hinge locking structure **330**. For example, the hinge locking structure **330** in the space **321** may move toward the shaft **310**. The hinge locking structure **330** may move to the outside of the shaft holder **320** through an opening **322** facing the shaft **310**. A stopper **323** of the shaft holder **320** may limit movement of the hinge locking structure **330** in the first direction (e.g., the z axis direction) opposite to the second direction (e.g., the -z axis direction) toward which the opening **322** is directed. For example, the seating space **321** may include the stopper **323** formed on an inner surface. The stopper **323** may include a stepped portion of the inner surface of the seating space **321**. As the hinge locking structure **330** moves in the first direction (e.g., the z axis direction), the hinge locking structure **330** may contact the stopper **323**. As the hinge locking structure **330** contacts the stopper **323**, movement in the first direction (e.g., the z axis direction) may be limited.

[0069] The hinge locking structure **330** may include a pin **331** and a head **332**. The pin **331** may be movable from the inside of the seating space **321** to the outside through the opening **322**. The pin **331** may be formed in a cylindrical shape, but is not limited thereto. The pin **331** may have various cross-sections and may have a length in the first direction (e.g., the z axis direction) or the second direction (e.g., the -z axis direction). The head **332** may be in contact with one end of the pin **331**. For example, the head **332** may be in contact with one end of the pin **331** facing the first direction (e.g., the z axis direction). For example, the head **332** may be in contact with one end facing the flexible display **240**. For example, the pin **331** may be extended from one surface of the head **332** in the second direction (e.g., the -z axis direction).

[0070] The hinge locking structure **330** may be in contact with the second magnet **340**. The second magnet **340** may be attached to a portion of the hinge locking structure **330**. The second magnet **340** may be disposed on one surface of the head **332** of the hinge locking structure **330**. The second magnet **340** may face the flexible display **240**. The second magnet **340** may interact with the first magnet **201**. For example, in the first state **200a** and the second state **200b**, the distance between the first magnet **201** and the second magnet **340** may be far. Within a designated distance, the first magnet **201** and the second magnet **340** may move the hinge locking structure **330** in the second direction (e.g., the -z axis direction) due to the repulsive force. For example, within the abnormal state of FIGS. **4A** and **4B** to be described later, the distance between the first magnet **201** and the second magnet **340** may be within a designated distance. Through the repulsive force of the first magnet **201** and the second magnet **340** disposed within the designated distance, the hinge locking structure **330** may move in the second direction (e.g., the -z axis direction).

[0071] The electronic device **200** may further include an elastic member **350**. The elastic member **350** may be disposed between the head **332** and the opening **322**. The elastic member **350** may be in contact with the head **332**. The elastic member **350** may be in contact with the inner surface of the seating space **321** in which the opening **322** is formed.

[0072] The elastic member **350** may elastically support the hinge locking structure **330** in the seating space **321**. The elastic member **350** may be disposed within the seating space **321** to surround the pin **331** of the hinge locking structure **330**. For example, the elastic member **350** may surround the outer circumferential surface of the pin **331**. For example, the elastic member **350** may be a spring. However, the elastic member **350** is not limited thereto, and may include a rubber or porous material. For example, the elastic member **350** may be compressed by an external force. The elastic member **350** may be restored when the external force is removed.

[0073] The elastic member **350** may be compressed by the first magnet **201** and the second magnet **340** being within the designated distance. Through the interaction of the first magnet **201** and the second magnet **340** (e.g., repulsive force), the hinge locking structure **330** coupled to the second magnet **340** may move in a direction away from the first magnet **201** (e.g., $-z$ axis direction).

[0074] The shaft **310** may include a recessed groove **360** formed on a part of the outer circumferential surface of the shaft **310** and a protrusion **370** disposed in the groove **360**. At least a part of the groove **360** may be overlapped with the hinge locking structure **330** along the outer circumferential surface of the shaft **310**, when the second magnet **340** is viewed from above. For example, the hinge locking structure **330** may be disposed on the groove **360**. The hinge locking structure **330** may be spaced apart from the groove **360** by an elastic force caused by the elastic member **350**, or be accommodated in at least a part of the space of the groove **360** through the interaction between the first magnet **201** and the second magnet **340**.

[0075] The groove **360** may include stoppers **360a**, **360b** that limit the rotation of the shaft **310** while at least a part of the pin **331** is inserted into the groove **360**. The stoppers **360a** and **360b** may be sidewalls of the groove **360**. For example, the stoppers **360a** and **360b** may be sidewalls extending substantially vertically from one surface of the groove **360**. The first stopper **360a** and the second stopper **360b** may be disposed in the periphery of the groove **360** formed by being recessed from the outer circumferential surface of the shaft **310** toward the one surface of the groove **360**. The first stopper **360a** may limit clockwise rotation of the shaft **310**. For example, while the pin **331** is inserted into the groove **360**, the pin **331** and the first stopper **360a** may be contacted through clockwise rotation of the shaft **310**. The shaft **310** cannot rotate clockwise due to the contact between the pin **331** and the first stopper **360a**. The second stopper **360b** may limit counterclockwise rotation of the shaft **310**. For example, while the pin **331** is inserted into the groove **360**, the pin **331** and the second stopper **360b** may be contacted through counterclockwise rotation of the shaft **310**. The shaft **310** cannot rotate counterclockwise due to the contact between the pin **331** and the second stopper **360b**.

[0076] The groove **360** may include a plurality of protrusions **370**. The plurality of protrusions **370** may protrude toward the outer circumferential surface of the shaft **310** within the groove **360**. For example, the plurality of protrusions **370** may protrude from one surface of the groove **360** in a radial direction. One surface of the groove **360** may be a surface facing the radial direction of the shaft **310**. The plurality of protrusions **370** may provide a plurality of accumulating spaces **361**, **362**, and **363** into which the pin **331** is insertable. The plurality of protrusions **370** may be spaced apart from each other by a designated interval. When the pin **331** is a polygon, the designated interval may correspond to the width of the pin **311**. For example, the designated interval may correspond to a width in a rotation direction of the shaft **310** of the pin **311**. When the pin **331** is circular, the designated interval may correspond to a diameter of an end surface of the pin **331**. The designated interval may be related to the size of the accommodating space **362** between the plurality of protrusions **370** or the size of the accommodating spaces **361** and **363** between one of the protrusions and the sidewall of the groove **360**. The designated interval may be an interval in

which the pin **311** is insertable into the accommodating spaces **361**, **362**, and **363**. For example, the designated interval may be equal to or larger than the width or diameter of the pin **311**. Although the pin **311** has been described as being polygonal or circular, it is not limited thereto. The cross-section of the pin **311** may have an oval shape, or may have a shape, which is insertable into the groove **360** or the accommodating spaces **361**, **362**, and **363**.

[0077] The plurality of protrusions **370** may include a first protrusion **370a** and a second protrusion **370b**. The plurality of protrusions **370** may provide a plurality of accommodating spaces **361**, **362**, and **363** into which the pin **331** is insertable. For example, the first protrusion **370a** and the first stopper **360a** may provide a first space **361**. The first space **361** may be overlapped with the pin **331** when viewed in the z axis direction, since the third housing **230** rotates with respect to the first housing **210** in the first state **200a**. Since the rotation angle of the third housing **230** with respect to the first housing **210** is limited by providing the first stopper **360a**, the first space **361** may prevent damage to the display in an abnormal state. The abnormal state will be described in detail below with respect to FIGS. **4A**, **4B**, **5A**, and **5B**.

[0078] The first protrusion **370a** and the second protrusion **370b** may provide a second space **362**. The second space **362** may be overlapped with the pin **331**, as the third housing **230** rotates with respect to the first housing **210** from the first state **200a**. The second protrusion **370b** and the second stopper **360b** may provide the third space **363**. The third space **363** may correspond to the position of the pin **331** in the first state **200a**. The pin **331** may be accommodated in the third space **363** by an interaction between the first magnet **201** and the second magnet **340**.

[0079] The plurality of protrusions **370** include the first protrusion **370a** and the second protrusion **370b**, and have been described as two protrusions, but this is only an example. The plurality of protrusions **370** may include one protrusion or may further include three or more protrusions, corresponding to the width of the groove **360**. According to an embodiment, the groove **360** may not include a plurality of protrusions. For example, the width of the groove **360** may have a size corresponding to the diameter of the pin **331** or the width of the pin **331**, and may not include protrusions. For example, in a case that the groove **360** has a width larger than the diameter of the pin **331** or the width of the pin **331**, the shaft **310** may rotate between the first stopper **360a** and the second stopper **360b** of the groove **360**.

[0080] The plurality of protrusions **370** and the side surface of the groove **360** may be parallel to the radial direction of the shaft. The side surface of the plurality of protrusions **370** may incline with respect to a radial direction of the shaft. The side surface having the inclinations of the plurality of protrusions **370** may allow a pin located in one of the spaces **361**, **362**, and **363** to move to another space. According to the movement of the pin **331**, the plurality of protrusions **370** may provide physical or tactile notifications to the user of the electronic device **200**. For example, the pin **331** of the hinge locking structure **330** may move along the side surface of the second protrusion **370b** having an inclination from one (e.g., the third accommodating space **363**) of the accommodating spaces **361**, **362**, and **363**. The pin **331** may move to an accommodating space (e.g., the second accommodating space **362**) next (adjacent) to the one accommodating space **363**, through the movement in the z axis direction along the side surface of the second protrusion **370b**.

[0081] FIG. **4A** illustrates an example of arrangement of a hinge locking structure in a third state of an example electronic device according to various embodiments. FIG. **4B** illustrates an example hinge locking structure in a third state of an example electronic device according to various embodiments. Referring to FIG. **4A**, the first hinge structure **202** may rotatably connect one side surface **221** of the second housing **220** facing the first side surface **211** of the first housing **210** and the first housing **210**. The second hinge structure **203** may rotatably connect one side surface **231** of the third housing **230** facing the second side surface **212** opposite to the first side surface **211** and the first housing **210**.

[0082] In the third state **200c** of the electronic device **200**, the first housing **210** and the second housing **220** may be folded, and the first housing **210** and the third housing **230** may be unfolded.

The third state **200c** may be a state in which the second housing **220** rotates with respect to the first housing **210** prior to rotation of the third housing **230** with respect to the first housing **210**. Although the third housing **230** should preferentially rotate with respect to the first housing **210** during the folding operation of the electronic device **200**, in the side surface where the second housing **220** preferentially rotates with respect to the first housing **210**, the third state **200c** may be referred to as an abnormal state. For example, in the third state **200c**, an angle formed by the second housing **220** and the first housing **210** may be an angle- $\theta 1'$. The third state **200c** may be a state in which the second housing **220** is further rotated by an angle $\theta 1'$ in the clockwise direction after the second housing **220** rotates 180 degrees clockwise in an unfolded state. In the abnormal state of the electronic device, since the third housing **230** cannot be folded to face the first housing **210**, the electronic device may be damaged when pressed to switch to the folded state. The flexible display **240** may be disposed on a curved surface or a planar surface formed by the first housing **210**, the second housing **220**, and the third housing **230**. For example, the flexible display **240** disposed on the first housing **210** and the second housing **220** may include a planar area and a curved area. The first display area **240a** disposed on the first housing **210** and the second display area **240b** disposed on the second housing **220** may be a planar area. The fourth display area **240d** disposed between the first display area **240a** and the second display area **240b** may include a curved area. The fourth display area **240d** may be bent according to folding of the first housing **210** and the second housing **220** by the first hinge structure **202**. The first display area **240a**, the third display area **240c**, and the fifth display area **240e** of the flexible display **240** disposed on the first housing **210** and the third housing **230** may be substantially form a planar surface. For example, since rotation of the third housing **220** according to the second hinge structure **203** does not exist, the first housing **210** and the third housing **230** may be identical to the first state **200a**.

[0083] When converted from the first state **200a** (e.g., the first state **200a** in FIG. 2A) to the second state **200b** (e.g., the second state **200b** in FIG. 2B), the third housing **230** may be folded with respect to the first housing **210**, before the second housing **220** is folded with respect to the first housing **210**. The abnormal state of the electronic device may be a state in which the device is folded differently from the above-described folding order. For example, in the third state **200c**, the second housing **220** may be folded to the first housing **210** before the third housing **230** is folded to the first housing **210**. For example, the second display area **240b** disposed on the second housing **220** may face the first display area **240a** disposed on the first housing **210**. The first magnet **201** adjacent to another side surface **222** of the second housing **220** may face the second hinge structure **203**. The first magnet **201** may interact with the magnetic body in the second hinge structure **203**. Within the third state **200c** in which a part of the surface of the first housing **210** contacts the second surface **220a** of the second housing **220**, the hinge locking structure **330** in the second hinge structure **203** may face the first magnet **201**. The operation of the hinge locking structure by the interaction between the first magnet **201** and the magnetic body (e.g., the second magnet **340**) in the second hinge structure **203** will be described in FIG. 4B.

[0084] Referring to FIG. 4B, the electronic device **200** may further include a hinge locking structure **330**. The hinge locking structure **330** may include the pin **331**. In the third state **200c**, the hinge locking structure **330** may be configured to move away from the first magnet **201** by the repulsive force between the first magnet **201** and the second magnet **340** within the seating space **321**, and insert the pin **331** into the groove **360**. The pin **331** may be inserted into the groove **360** formed in at least one shaft **310** within the second hinge structure **203**. In the third state **200c**, the pin **331** may protrude to the outside of the shaft holder **320** through the opening **322** by the interaction of the first magnet **201** and the second magnet **340** (e.g., repulsive force). In the third state **200c**, the polarity of the first magnet **201** may be the same as the polarity of the second magnet **340**. In the third state **200c**, the first magnet **201** and the second magnet **340** may provide pushing force to each other by the interaction. For example, the first magnet **201** may provide a repulsive force (pushing the second magnet **340**) to the second magnet **340**. The second magnet

340 may provide a repulsive force pushing the first magnet **201** to the first magnet **201**. Since the first magnet **201** is fixed to the inner space close to the other side surface **222** of the second housing **220**, the hinge locking structure **330** may be moved from the inside of the seating space **321** to the outside by the force transmitted to the second magnet **340**. For example, the hinge locking structure **330** may move the pin **331** to the inside of the seating space **321** through the second magnet **340** attached (or connected) to the head **332** of the hinge locking structure **330**. The pin **331** protruding to the outside of the shaft holder **320** may occupy at least a part of the groove **360** of the shaft **310**. For example, the pin **331** may be inserted into the third space **363** located under the hinge locking structure **330** in the third state **200c**. The pin **331** inserted into the third space **363** of the groove **360** may limit rotation of the third housing **230** with respect to the first housing **210**. In a state in which the pin **331** is inserted into the third space **363**, rotation of the shaft **310** may be limited. When the shaft **310** rotates counterclockwise, the pin **331** and the second stopper **360b** may contact each other, and thus rotation may be limited. When the shaft **310** rotates clockwise, the pin **331** and the second protrusion **370b** may contact each other, and thus rotation may be limited. The pin **331** may ride over the second protrusion **370b** when the second protrusion **370b** has an inclined surface, but may give a physical notification to the user.

[0085] While the hinge locking structure **330** is switched from the third state **200c** to the first state **200a**, the pin **331** may move from the groove **360** to the seating space **321** by the elastic member **350**. For example, when converted to the first state **200a**, the distance between the first magnet **201** and the second magnet **340** may increase. As the repulsive force applied to the second magnet **340** is weakened, the elastic force of the elastic member **350** may be greater than the repulsive force. The elastic member **350** may push up the head **332**. As the head **332** moves, the pin **331** integrally formed with the head **332** may move into the seating space **321**. According to the movement of the head **332**, the pin **331** integrally formed with the head **332** may move into the seating space **321**.

[0086] According to the above-described embodiment, in the third state **200c**, the electronic device **200** may limit the movement of the third housing **230** or may provide a notification to the user when the third housing **230** moves.

[0087] FIG. 5A illustrates an example of arrangement of a hinge locking structure when a third housing moves in a fourth state of an example electronic device according to various embodiments. FIG. 5B illustrates an example hinge locking structure when a third housing moves in a fourth state of an example electronic device according to various embodiments.

[0088] Referring to FIG. 5A, in the third state **200c** (e.g., the third state **200c** of FIG. 4A) of the electronic device **200**, the third housing **230** may rotate with respect to the first housing **210** through the second hinge structure **203**. The fourth state **200d** may be one of the abnormal states of the electronic device. For example, the abnormal state may include the third state **200c** in which the second housing **220** contacts a part of the first housing **210**, a state of switching from the third state **200c** to the fourth state, and a fourth state **200d** in which a part of the third housing **230** contacts the side surface of the second housing **220** while the second housing **220** contacts a part of the first housing **210**.

[0089] Even within the fourth state **200d**, positions of the first housing **210** and the second housing **220** may be the same as or similar to those of the third state **200c**.

[0090] In the electronic device **200** in the fourth state **200d**, the first housing **210** and the second housing **220** may be folded, and the first housing **210** and the third housing **230** may be folded. The third housing **230** may be in contact with a part of the second housing **220**. The fourth state **200d** may be a state in which the third housing **230** rotates counterclockwise by a third angle θ_3 from the third state **200c**. When the third housing **230** rotates counterclockwise further than the third angle θ_3 so that the third housing **230** faces the first housing **210**, the flexible display **240** or the electronic device **200** may be damaged.

[0091] In the fourth state **200d**, the first display area **240a** and the second display area **240b** of the flexible display **240** may face each other. The second housing **220** may contact the fifth display

area **240e**. In the fourth state **200d**, the angle of the third display area **200c** with the first display area **200a** may be difficult to be further reduced. For example, the third housing **230** may be movable within the designated angular range to the first housing **210**, within the third state **200c** among the abnormal states.

[0092] The designated angular range may be up to approximately 180 degrees from the angle $\theta 2'$ between the first housing **210** and the third housing **230** in the fourth state **200d**. For example, in the fourth state **200d**, the angle $\theta 2'$ between the first housing **210** and the third housing **230** adjacent to the second housing **220** may be related to the position where the third housing **230** can no longer rotate counterclockwise. The angle at which the third housing **230** is moveable in the abnormal state may be from an angle $\theta 2'$ between the first surface **210a** and the third surface **230a** adjacent to the second housing **220** in the fourth state **200d** in which the third surface **230a** is adjacent to the second housing **220**, to an angle $\theta 2$ (e.g., approximately 180 degrees) between the first surface **210a** and the third surface **230a**, for example, in the first state **200a** or the third state **200c**.

[0093] The flexible display **240** may be disposed on a curved surface or a planar surface formed by the first housing **210**, the second housing **220**, and the third housing **230**. For example, the flexible display **240** disposed on the first housing **210** and the second housing **220** may include a planar area and a curved area. The first display area **240a** disposed on the first housing **210** and the second display area **240b** disposed on the second housing **210** may be planar areas. The fourth display area **240d** disposed between the first display area **240a** and the second display area **240b** may include a curved area. The fourth display area **240d** may be bent by the first hinge structure **202**, according to the folding of the first housing **210** and the second housing **220**. The first display area **240a**, the third display area **240c**, and the fifth display area **240e** of the flexible display **240** disposed on the first housing **210** and the third housing **230** may include a planar area and a curved area. The first display area **240a** disposed on the first housing **210** and the third display area **240c** disposed on the third housing **230** may be a planar area. The fifth display area **240e** disposed between the first display area **240a** and the second display area **240b** may include a curved area. The fifth display area **240e** may be bent by the second hinge structure **203**, according to the folding of the first housing **210** and the third housing **230**.

[0094] The first magnet **201** may interact with a magnetic body (e.g., the second magnet **340**) in the second hinge structure **203**. In the third state **200c** in which a part of the first surface **210** of the first housing is adjacent to the second surface **220a** of the second housing (**220**), the hinge locking structure **330** within the second hinge structure **203** may face the first magnet **201**. The operation of the hinge locking structure **330** by the interaction of the magnetic body within the first magnet **201** and the second hinge structure **203** will be described in FIG. 5B.

[0095] Referring to FIG. 5B, the hinge locking structure **330** may operate within the influence of the reactive force acting between the first magnet **201** and the second magnet **340** even while switching from the third state **200c** to the fourth state **200d**. In the third state **200c**, when the third housing **230** moves, the pin **331** may move from one space of the accommodating spaces **361**, **362**, and **363** to another space in contact with the one protrusion, by moving along the surface of one of the plurality of protrusions **370**. In the fourth state **200d**, the pin **331** may be accommodated in the first space **361**. The pin **331** may be in contact with the first stopper **360a**. The hinge locking structure **330** may disengage the pin **331** of the hinge locking structure **330** from the groove **360** by the elastic member **350**, by deviating from the third state **200c** or the fourth state **200d**, which is the abnormal state. For example, the shaft **310** from which the pin **331** is removed may be switched to a first state **200a**, which is an unfolded state. The rotation of the shaft **310** in the clockwise direction may be limited by the pin **331** in contact with the first stopper **360a**. In the fourth state **200d**, while the second housing **220** rotates counterclockwise from the first housing **210**, the repulsive force acting between the first magnet **201** and the second magnet **340** may be changed in the hinge locking structure **330**. For example, the elastic member **350** may be changed based on a

change in the repulsive force. The pin **331** may move from the groove **360** (e.g., the first space **361**) into the seating space **321** by the elastic member **350**.

[0096] According to the above-described embodiment, the hinge locking structure **330** may reduce damage to the flexible display **240** and the electronic device **200** within the fourth state **200d**, by limiting the counterclockwise rotation of the third housing **230**. For example, in the hinge locking structure **330**, as the elastic member **350** is compressed, the pin **331** may move outside the shaft holder **320**. The pin **331** moved to the outside of the shaft holder **320** may occupy one of the grooves **360** or a plurality of spaces **361**, **362**, and **363** in the groove **360**. The hinge locking structure **330** may be configured to operate in an abnormal state (e.g., the third state **200c** or the fourth state **200d**) of the electronic device. The electronic device **200** may limit the rotation of the shaft **310** through the hinge locking structure **330** within the abnormal state. The electronic device **200** may prevent damage to the flexible display **240** and the electronic device **200** using the shaft **310** having limited rotation.

[0097] FIG. **6** is a block diagram of an example electronic device according to various embodiments.

[0098] Referring to FIG. **6**, the electronic device **600** (e.g., the electronic device **101** of FIG. **1** or the electronic device **200** of FIG. **2A**) may include a processor **620** (e.g., the processor **120** of FIG. **1**), sensors **641** and **646** (e.g., the sensor module **176** of FIG. **1**), and an output device **660** (e.g., the sound output module **155** of FIG. **1** or the display module **160**).

[0099] The electronic device **600** may control the first sensor **641**, the second sensor **646**, and the output device **660** through the processor **620**. The processor **620** may obtain data on the state of the electronic device **600** through the first sensor **641** and the second sensor **646**.

[0100] The processor **620** may be electrically connected to the first sensor **641**. The processor **620** may obtain first data related to the posture of the second housing (e.g., the second housing **220** of FIG. **2A**) through the first sensor **641**. The first sensor **641** may be a sensor capable of obtaining sensing data related to the posture of the second housing **220**. For example, the first sensor **641** may obtain data related to an angle between the first housing **210** and the second housing **220**.

[0101] The processor **620** may be electrically connected to the second sensor **646**. The processor **620** may obtain second data related to the posture of the third housing (e.g., the third housing **230** of FIG. **2A**) through the second sensor **646**. The second sensor **646** may be a sensor capable of obtaining sensing data related to the posture of the third housing **230**. For example, the second sensor **646** may obtain data related to an angle between the first housing **210** and the third housing **230**.

[0102] The first sensor **641** and the second sensor **646** may obtain data related to the distance, angle, posture sensed, and/or changes in magnetic field. Based on the distance, angle, posture, and/or the change in magnetic field, the processor **620** may identify an angle between the first housing **210** and the second housing **220** and/or an angle between the first housing **210** and the third housing **230**. For example, the first sensor **641** may be disposed in the first housing **210** and/or the second housing **220**. The first sensor **641** may include, for example, a proximity sensor, a gyro sensor, or a hall sensor. For example, when disposed in the first housing **210**, the first sensor **641**, which is a proximity sensor or a hall sensor, may obtain data related to a distance between the first housing **210** and the second housing **220**. Although the first sensor **641** is described as being disposed in the first housing **210**, the first sensor **641** may also be disposed in the second housing **220**. The processor **620** may identify an angle between the first housing **210** and the second housing **220** based on the data related to the distance. For example, the first sensor **641**, which is a gyro sensor or an acceleration sensor, may be disposed in the second housing **220** rotating based on the first housing **210**. The first sensor **641** may identify the posture of the second housing **220**. Based on the identified posture of the second housing **220**, the processor **620** may identify an angle between the first housing **210** and the second housing **220**.

[0103] The second sensor **646** may include, for example, a proximity sensor, a gyro sensor, or a

hall sensor. For example, when disposed in the first housing **210**, the second sensor **646**, which is a proximity sensor or a hall sensor, may obtain data related to the distance between the first housing **210** and the third housing **230**. Although the second sensor **646** is described as being disposed in the first housing **210**, the second sensor **646** may also be disposed in the third housing **230**. The processor **620** may identify an angle between the first housing **210** and the third housing **230** based on the data related to the distance. For example, the second sensor **646**, which is a gyro sensor or an acceleration sensor, may be disposed in the third housing **230** rotating based on the first housing **210**. The first sensor **641** may identify the posture of the third housing **230**. Based on the identified posture of the third housing **230**, the processor **620** may identify an angle between the first housing **210** and the third housing **230**.

[0104] Based on the first data obtained through the first sensor **641** and the second data obtained through the second sensor **646**, the processor **620** may identify the state of the first housing **220** or the third housing **230** based on the first housing **210**. For example, the processor **620** may identify the first angle θ_1 between the first housing **210** and the second housing **220** or the second angle θ_2 between the first housing **210** and the third housing **230**, based on the first data and the second data. For example, the processor **620** may identify that the electronic device **200** is in the first state **200a** based on identifying that the first angle θ_1 is approximately 180 degrees. The processor **620** may identify that the electronic device **200** is in the second state **200b** based on identifying that the first angle θ_1 is approximately 0 degrees. For example, the processor **620** may identify that the electronic device **200** is in the first state **200a** based on identifying that the second angle θ_2 is approximately 180 degrees. The processor **620** may identify that the electronic device **200** is in the second state **200b** based on identifying that the second angle θ_2 is approximately 0 degrees.

According to the state change of the electronic device **200**, the first angle θ_1 and the second angle θ_2 may change. For example, as the electronic device **200** changes from the first state **200a** to the second state **200b**, the first angle θ_1 and the second angle θ_2 may change to approximately 0 degrees from 180 degrees.

[0105] The processor **620** may be configured to provide a first notification to the outside through output device **660**, based on the first angle θ_1 between the first housing **210** and the second housing **220** and the second angle θ_2 between the first housing **210** and the third housing **230**. The output device **660** may include a display module **160** or a sound output module **155**. The display module **160** may visually provide a notification based on the first angle θ_1 and the second angle θ_2 . The sound output module **155** may audibly provide a notification based on the first angle θ_1 and the second angle θ_2 .

[0106] The processor **620** may identify the abnormal state based on the first angle θ_1 being less than the second angle θ_2 . The abnormal state may be a state in which the postures of the first housing **210**, the second housing **220**, and the third housing **230** are different from previously designated postures. The designated postures may be postures in which the second housing **220** is foldable after the third housing **230**. The first hinge structure (e.g., the first hinge structure **202** of FIG. 2A) rotatably connecting the second housing **220** to the first housing **210** may have a wider width than the second hinge structure (e.g., the second hinge structure **203** of FIG. 2A) rotatably connecting the third housing **230** to the first housing **210**. Due to the wide width of the first hinge structure **202**, the second housing **220** may be folded after the third housing **230** and changed to the second state **200b** (e.g., the second state **200b** of FIG. 2B). In order to be a fully folded second state (e.g., the second state **200b** of FIG. 2B), the third housing **230** may be folded on the first housing **210**, and the second housing **220** may be disposed on the folded third housing **230** (e.g., the +z axis direction). In a case that the electronic device **600** cannot be in a fully folded state (e.g., the second state **200b**) based on the first angle θ_1 and the second angle θ_2 , the processor **620** may provide a first notification to the outside through the output device **660**.

[0107] The processor **620** may provide another second notification distinguished from the first notification to the outside in response to identifying that the first angle θ_1 is less than or equal to 0

degrees. The processor **620** may identify that the first angle θ_1 is less than or equal to 0° while the first angle θ_1 is less than the second angle θ_2 . A case in which the first angle θ_1 is smaller than the second angle θ_2 within the designated angular range of the first angle θ_1 and the second angle θ_2 may be an abnormal state of the electronic device **600**. For example, in a case that the first angle θ_1 is smaller than the second angle θ_2 within the range of the first angle θ_1 and the second angle θ_2 from 0 degrees to 90 degrees, the state of the electronic device **600** may be an abnormal state. When the first angle θ_1 is 0 degrees or the first angle θ_1 is less than 0 degrees, the state of the electronic device **600** may be a third state **200c** (e.g., the third state **200c** of FIG. 4A). When the second angle θ_2 between the third housing **230** and the first housing is less than the designated angle by operating the third housing **230**, while a state of the electronic device **600** is the third state **200c**, the flexible display (e.g., the flexible display **240** of FIG. 2A) or the electronic device **600** may be damaged. The processor **620** may provide a notification related to the possibility of damage to the flexible display **240** or the electronic device **600** to the user through the output device **660**. The processor **620** may identify that the second angle θ_2 between the third housing **230** and the first housing **210** is less than or equal to the designated angle. The processor **620** may be configured to provide a third notification to the outside based on the identification that the second angle θ_2 is less than or equal to the designated angle.

[0108] The electronic device **600** may identify an abnormal state of the electronic device by identifying the first angle θ_1 between the first housing **210** and the second housing **220** and the second angle θ_2 between the first housing **210** and the third housing **230**. The processor **620** may provide a different notification to the user according to the degree of the abnormal state based on the identified abnormal state. The user may stop the folding operation by recognizing the possibility of damage to the display or the electronic device through the provided notification.

[0109] Considering that the flexible display **240** is covered by the housings **210**, **220**, and **230** in the second state **200b**, the electronic device **200** may be referred to as an in-folding electronic device. The electronic device **200** of the above-described embodiment has been described based on the in-folding electronic device, but is not limited thereto. The electronic device may be an out-folding electronic device in which the flexible display **240** is disposed on the outer surface of the electronic device and exposed in the folded state. For example, the first angle θ_1 and the second angle θ_2 described above may vary within a range of 180 degrees to 360 degrees. Referring back to FIG. 2A, in the unfolded state, the second housing **220** may rotate counterclockwise based on the first housing **210**, and the third housing **230** may rotate clockwise based on the first housing **210**.

[0110] The second housing **220** may be folded with respect to the first housing **210**, based on the hinge structure or the width of the hinge cover being different, after the third housing **230** is folded with respect to the first housing **210** similar to the above-described embodiment. In order to prevent damage when the second housing **220** is preferentially folded, the third magnet (not illustrated) and the fourth magnet (not illustrated) may be disposed at the same/similar positions to the arrangement positions of the first magnet **201** and the second magnet **340**.

[0111] FIG. 7 is a flowchart illustrating an example operation of detecting an abnormal state of an electronic device according to various embodiments.

[0112] Referring to FIG. 7, in operation **710**, the processor **620** (e.g., the processor **620** of FIG. 6) may obtain the first angle θ_1 and the second angle θ_2 based on the first data and the second data sensed through the first sensor (e.g., the first sensor **641** in FIG. 6) and the second sensor (e.g., the second sensor **646** in FIG. 6). The first data may be a distance or an angle between the first housing **210** and the second housing **220** obtained through the first sensor **641**. The second data may be a distance or an angle between the first housing **210** and the third housing **230** obtained through the second sensor **646**.

[0113] In operation **720**, the processor **620** may provide a notification to the outside based on the first angle θ_1 and the second angle θ_2 . The processor **620** may identify the state of the electronic device based on the first angle θ_1 and the second angle θ_2 . The processor **620** may provide various

notifications to the outside based on the state of the electronic device. For example, in response to identifying that the state of the electronic device is the abnormal state, a first notification related to the abnormal state of the electronic device may be provided externally. For example, in a case that the state of the electronic device is identified as the third state (e.g., the third state **200c** of FIG. 4A), the processor **620** may provide a second notification to warn that the display or electronic device may be damaged when the third housing is moved more than the designated angle. For example, in a case that the state of the electronic device is identified as the fourth state (e.g., the fourth state **200d** of FIG. 4A), the processor **620** may provide a third notification to warn that the display or electronic device may be damaged when the third housing is further moved.

[0114] In FIGS. 6 and 7, the electronic device may be configured to prevent damage to the electronic device or the display by notifying the abnormal state of the electronic device by identifying the first angle $\theta 1$ and the second angle $\theta 2$. The electronic device may physically limit the movement of the third housing **230** in the abnormal state (e.g., the third state **200c** or the fourth state **200d**) by having a hinge structure including the hinge locking structure **330** of FIGS. 3A to 5B.

[0115] According to the above-described example embodiments, the electronic device may reduce the possibility of damage to the electronic device due to movement of the third housing, by providing a locking structure that physically limits the movement of the third housing when the abnormal state is reached. The electronic device may notify the wrong (potentially damaging) movement of the third housing, by alerting the user by providing a step-by-step notification of abnormal state.

[0116] According to the above-described example embodiments, an electronic device (e.g., the electronic device **101** of FIG. 1 or the electronic device **200** of FIG. 2A) may include a first housing (e.g., the first housing **210** of FIG. 2A) including a first surface (e.g., the first surface **210a** of FIG. 2A), a second housing (e.g., the second housing **220** of FIG. 2A) including a second surface (e.g., the second surface **220a** of FIG. 2A), and a third housing (e.g., the third housing **230** of FIG. 2A) including a third surface (e.g., the third surface **230a** of FIG. 2A).

[0117] According to an example embodiment, the electronic device may include a first hinge structure (e.g., the first hinge structure **202** of FIG. 2A) and a second hinge structure (e.g., the second hinge structure **260** of FIG. 2A). The first hinge structure may rotatably connect one side surface (e.g., one side surface **221** of FIG. 3A) of the second housing facing a first side surface (e.g., the first side surface **211** of FIG. 3A) of the first housing to the first housing. The second hinge structure may rotatably connect one side surface (e.g., one side surface **231** of FIG. 3A) of the third housing facing a second side surface (e.g., the second side surface **212** of FIG. 3A) of the first housing opposite to the first side surface to the first housing. The electronic device may further include a flexible display (e.g., the flexible display **240** of FIG. 2A). The flexible display may be disposed on a surface formed by the first housing, the second housing, and the third housing. According to an example embodiment, the electronic device may further include a first magnet (e.g., the first magnet **201** of FIG. 3A). The first magnet may be disposed in the second housing. According to an example embodiment, the electronic device may further include a hinge locking structure (e.g., the hinge locking structure **330** of FIG. 3B). The hinge locking structure may include a pin (e.g., the pin **331** of FIG. 3B) facing the first magnet and insertable into a groove formed in at least one shaft in the second hinge structure (e.g., the groove **360** of FIG. 3B), in an abnormal state of the electronic device in which a part of the first surface is adjacent to the second surface.

[0118] According to an example embodiment, the first hinge structure and the second hinge structure may switch the electronic device into an unfolded state in which the first surface, the second surface, and the third surface face the same direction, or a folded state in which at least one of the first surface, the second surface, and the third surface face different directions.

[0119] According to an example embodiment, the pin may be spaced apart (or disengaged) from

the groove in the hinge locking structure as the abnormal state is changed to the unfolded state.

[0120] According to an example embodiment, the first magnet may be disposed in a periphery portion (e.g., the periphery portion **229** of FIG. **3A**) of the second housing including the other side surface of the second housing facing the one side surface of the second housing.

[0121] According to an example embodiment, the hinge locking structure may include a second magnet (e.g., the second magnet **340** of FIG. **3B**). The second magnet may be disposed on a surface of the hinge structure facing the flexible display.

[0122] According to an example embodiment, the polarity of the first magnet may be the same as that of the second magnet.

[0123] According to an example embodiment, in the abnormal state of the electronic device, the hinge locking structure may be configured to insert the pin into the groove by moving from the first magnet within the seating space by the repulsive force between the first magnet and the second magnet.

[0124] According to an example embodiment, the second hinge structure may further include a shaft holder (e.g., the shaft holder **320** of FIG. **3B**). The shaft holder **320** may rotatably support the at least one shaft.

[0125] According to an example embodiment, the shaft holder may include a seating space in which the hinge locking structure is seated. The shaft holder may include an elastic member (e.g., the elastic member **350** of FIG. **3B**). The elastic member may elastically support the hinge locking structure within the seating space.

[0126] According to an example embodiment, while the electronic device is changed to the unfolded state from the abnormal state, the pin may be moved from the groove to the seating space by the elastic member.

[0127] According to an example embodiment, the at least one shaft may rotate as the electronic device changes to a folded state or an unfolded state, and may be configured to rotate the third housing with respect to the first housing.

[0128] According to an example embodiment, the groove may include a plurality of protrusions (e.g., the plurality of protrusions **370** of FIG. **3B**) protruding in a radial direction of the shaft and spaced apart by a designated interval in the groove.

[0129] According to an example embodiment, the designated interval may correspond to the width (or other dimension) of the pin.

[0130] According to an example embodiment, the plurality of protrusions may provide a plurality of accommodating spaces into which the pin is insertable.

[0131] According to an example embodiment, a side surface of the groove may be parallel to the radial direction of the shaft.

[0132] According to an example embodiment, the side surface of the plurality of protrusions may have an inclination relative to a radial direction of the shaft.

[0133] According to an example embodiment, when the third housing moves within the abnormal state of the electronic device, the pin may move from one space (e.g., a third space **363**) of the accommodating spaces (e.g., the first space **361**, the second space **362**, and the third space **363**) to another space (e.g., a second space **362**) contacting the one protrusion by moving along the surface of one of the plurality of protrusions.

[0134] According to an example embodiment, the third housing may be movable within a designated angular range in the abnormal state of the electronic device with respect to the first housing.

[0135] According to an example embodiment, the designated angular range may be 180 degrees from an angle between the first surface contacting the second housing in the abnormal state and the third surface contacting the side surface of the second housing.

[0136] According to an example embodiment, in the unfolded state, an overlapped area of the display and the first hinge structure when the first hinge structure is viewed from above may be

wider than an overlapped area of the display and the second hinge structure when the second hinge structure is viewed from above. For example, the width of the first hinge cover may be wider than the width of the second hinge cover.

[0137] According to an example embodiment, the electronic device may include a processor (e.g., the processor **120** of FIG. **1** or the processor **620** of FIG. **6**), a first sensor (e.g., the first sensor **641** of FIG. **6**), and a second sensor (e.g., the second sensor **646** of FIG. **6**).

[0138] According to an example embodiment, the first sensor may obtain first data related to the posture of the second housing and may be disposed in the second housing.

[0139] According to an example embodiment, the second sensor may obtain second data related to the posture of the third housing and may be disposed in the third housing.

[0140] According to an example embodiment, the processor may obtain a positional relationship of the second housing or the third housing based on the first housing based on the first data and the second data sensed through the first sensor. The positional relationship may be expressed by an angle.

[0141] According to an example embodiment, the processor may be configured to provide a notification to the outside, in response to identifying that the first angle θ_1 between the first housing and the second housing is less than the second angle θ_2 between the first housing and the third housing.

[0142] According to an example embodiment, the processor may be configured to provide another notification distinguished from the notification to the outside in response to identifying that the first angle θ_1 is 0 degrees.

[0143] According to an example embodiment, the pin inserted into the groove may limit rotation of the third housing with respect to the first housing.

[0144] According to an example embodiment, the electronic device (e.g., the electronic device **101** of FIG. **1** or the electronic device **200** of FIG. **2A**) may include a first housing (e.g., the first housing **210** of FIG. **2A**) including a first surface (e.g., the first surface **210a** of FIG. **2A**), a second housing (e.g., the second housing **220** of FIG. **2A**) including a second surface (e.g., the second surface **220a** of FIG. **2A**), and a third housing (e.g., the third housing **230** of FIG. **2A**) including a third surface (e.g., the third surface **230a** of FIG. **2A**).

[0145] According to an example embodiment, the electronic device may include a first hinge structure (e.g., the first hinge structure **202** of FIG. **2A**) and a second hinge structure (e.g., the second hinge structure **203** of FIG. **2A**). The first hinge structure may rotatably connect one side surface (e.g., one side surface **221** of FIG. **3A**) of the second housing facing a first side surface (e.g., the first side surface **211** of FIG. **3A**) of the first housing to the first housing. The second hinge structure may rotatably connect one side surface (e.g., the side surface **231** of FIG. **3A**) of the third housing facing a second side surface (e.g., the second side surface **212** of FIG. **3A**) of the first housing opposite to the first side surface.

[0146] The example electronic device may further include a flexible display (e.g., the flexible display **240** of FIG. **2A**). The flexible display may be disposed on a surface formed by the first housing, the second housing, and the third housing.

[0147] According to an example embodiment, the electronic device may include a processor (e.g., the processor **120** of FIG. **1** or the processor **620** of FIG. **6**), a first sensor (e.g., the first sensor **641** of FIG. **6**), and a second sensor (e.g., the second sensor **646** of FIG. **6**). According to an example embodiment, the first sensor may obtain first data related to the posture of the second housing and may be disposed in the second housing. According to an example embodiment, the second sensor may obtain second data related to the posture of the third housing and may be disposed in the third housing.

[0148] According to an example embodiment, the electronic device may include a first magnet within the second housing.

[0149] According to an example embodiment, the electronic device may include a hinge locking

structure (e.g., the hinge locking structure **330** of FIG. **3B**). The hinge locking structure may include a pin (e.g., the pin **331** of FIG. **3B**) facing the first magnet and insertable into a groove formed in at least one shaft in the second hinge structure, in abnormal state of the electronic device in which a part of the first surface is adjacent to the second surface.

[0150] According to an example embodiment, the first hinge structure and the second hinge structure may be configured to provide an unfolded state in which the first surface, the second surface, and the third surface face the same direction, or a folded state in which at least one of the first surface, the second surface, or the third surface face different directions.

[0151] According to an example embodiment, an overlapped area between the display and the first hinge structure when the display is viewed from above in the unfolded state may be larger than an overlapped area between the display and the second hinge structure.

[0152] According to an example embodiment, the electronic device may further include a processor (e.g., the processor **620** of FIG. **6**).

[0153] According to an example embodiment, the processor may obtain the angle of the second housing or the third housing based on the first housing based on the first data and the second data obtained through the first sensor, and provide a first notification to the outside in response to identifying that the first angle θ_1 between the first housing and the second housing is smaller than the second angle θ_2 between the first housing and the third housing.

[0154] According to an example embodiment, the processor may be configured to provide a second notification distinguished from the first notification to the outside in response to identifying that the first angle θ_1 is 0 degrees.

[0155] According to an embodiment, the processor may be configured to identify that the second angle θ_2 is less than or equal to the designated angle and provide a third notification to the outside based on an identification that the second angle θ_2 is below the designated angle.

[0156] An example electronic device according to various embodiments may be one of various types of electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smartphone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, a home appliance, or the like. According to an embodiment of the disclosure, the electronic devices are not limited to those described above.

[0157] It should be appreciated that various embodiments of the present disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements. It is to be understood that a singular form of a noun corresponding to an item may include one or more of the things, unless the relevant context clearly indicates otherwise. As used herein, each of such phrases as “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B, or C,” “at least one of A, B, and C,” and “at least one of A, B, or C,” may include any one of, or all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as “1st” and “2nd,” or “first” and “second” may be used to simply distinguish a corresponding component from another, and do not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with,” “coupled to,” “connected with,” or “connected to” another element (e.g., a second element), the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element.

[0158] According to various embodiments, each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities, and some of the multiple entities may be separately disposed in different components. According to various embodiments, one or more of the above-described components may be omitted, or one or more other components may be added. Alternatively or additionally, a plurality of components (e.g.,

modules or programs) may be integrated into a single component. In such a case, according to various embodiments, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to various embodiments, operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added. [0159] While the disclosure has been illustrated and described with reference to various example embodiments, it will be understood that the various example embodiments are intended to be illustrative, not limiting. It will be further understood by those of ordinary skill in the art that various changes in form and detail may be made without departing from the true spirit and full scope of the disclosure, including the appended claims and their equivalents. It will also be understood that any of the embodiment(s) described herein may be used in conjunction with any other embodiment(s) described herein.

Claims

1. A foldable electronic device comprising: a first housing part; a second housing part; a third housing part; a first hinge rotatably coupling the first housing part and the second housing part; a second hinge rotatably coupling the first housing part and the third housing part; a flexible display disposed on a first surface of the first housing part, a second surface of the second housing part and a third surface of the third housing part; at least one sensor; at least one processor comprising processing circuitry; and memory comprising one or more storage media storing instructions, when executed by the at least one processor individually or collectively, cause the foldable electronic device to: using the at least one sensor, identify a folding state of the foldable electronic device, and based at least in part on the folding state of the foldable electronic device corresponding to an abnormal folded state, display a warning message related to the folding state via at least a portion of the flexible display.
2. The foldable electronic device of claim 1, wherein the instructions, when executed by the at least one processor individually or collectively, cause the foldable electronic device to, based on an angle between the first housing part and the second housing part while identifying a state in which the third housing part is folded with respect to the first housing part, identify the abnormal folded state.
3. The foldable electronic device of claim 1, wherein the at least one sensor includes a first sensor and a second sensor, and wherein the instructions, when executed by the at least one processor individually or collectively, cause the foldable electronic device to: using the first sensor, obtain a first data related to a first angle, a first distance, or first magnetic field between the first housing and the second housing, using the second sensor, obtain a second data related to a second angle, a second distance, or second magnetic field between the first housing and the third housing, and based at least in part on the first data and the second data, identify the folding state of the foldable electronic device.
4. The foldable electronic device of claim 1, wherein instructions, when executed by the at least one processor individually or collectively, cause the foldable electronic device to display the warning message related to the folding state when the foldable electronic device is in the abnormal folded state.
5. The foldable electronic device of claim 1, wherein the warning message includes an advice to a user of the foldable electronic device regarding a future folding state of the foldable device.
6. The foldable electronic device of claim 1, wherein instructions, when executed by the at least one processor individually or collectively, cause the foldable electronic device to display the warning message related to the folding state based on a determination that the foldable electronic

device may be damaged if the third housing is moved more than a designated angle.

7. The foldable electronic device of claim 1, wherein instructions, when executed by the at least one processor individually or collectively, cause the foldable electronic device to display the warning message related to the folding state based on a determination that the foldable electronic device may be damaged if the third housing is further moved.

8. The foldable electronic device of claim 1, wherein the at least one sensor includes at least one of a proximity sensor, a gyro sensor, a hall sensor or an acceleration sensor.

9. The foldable electronic device of claim 1, wherein the instructions, when executed by the at least one processor individually or collectively, cause the foldable electronic device to: while the first housing part and the second housing part are unfolded with respect to each other, identify, via the at least one sensor, a folding movement of the first and third housing parts to the abnormal folded state, and based on identifying the first and third housing parts being moved to the abnormal folded state, display, via the flexible display, a notification regarding the abnormal folded state of the foldable electronic device.

10. The foldable electronic device of claim 9, wherein the abnormal folded state is defined within a rotational range of the third housing part being rotated with respect to the first housing part.

11. The foldable electronic device of claim 9, wherein the folding movement of the first and third housing parts to the abnormal folded state is identified via the at least one sensor while the third housing part is rotated with respect to the first housing part within a rotational range allowed by the second hinge.

12. The foldable electronic device of claim 9, wherein the abnormal folded state is defined within a rotational range from a first state in which the first and third housing parts are unfolded with respect to each other to a second state in which the first and third housing parts are folded with respect to each other.

13. The foldable electronic device of claim 9, wherein the abnormal folded state of the foldable electronic device is defined while allowing the third housing part to rotate with respect to the first housing part via the second hinge assembly.

14. The foldable electronic device of claim 1, wherein the at least one sensor includes: at least one first sensor for identifying a folding movement of the first and third housing parts, and at least one second sensor for identifying a folding movement of the first and second housing parts, wherein the instructions, when executed by the at least one processor individually or collectively, cause the foldable electronic device to: identify an angle between the first housing part and the second housing part via the at least one second sensor, based on identifying that the angle between the first housing part and the second housing part is within a threshold angle range via the at least one second sensor, identify that the first housing part and the second housing part are folded, and based on identifying that the angle between the first housing part and the second housing part is out of the threshold angle range via the at least one second sensor, identify that the first housing part and the third housing part are unfolded.

15. The foldable electronic device of claim 14, wherein the threshold angle range is a first threshold angle range, and wherein the instructions, when executed by the at least one processor individually or collectively, cause the foldable electronic device to identify the third housing part at least partially folded with respect to the first housing part by identifying that an angle between the first housing part and the third housing part is within a second threshold angle range via the at least one first sensor.

16. The foldable electronic device of claim 1, wherein the at least one sensor includes: at least one first sensor for identifying a folding movement of the first and third housing parts, and at least one second sensor for identifying a folding movement of the first and second housing parts, wherein the instructions, when executed by the at least one processor individually or collectively, cause the foldable electronic device to, based on identifying that the first housing part and the second housing part are unfolded via the at least one second sensor, enable identifying whether the third housing

part is at least partially folded with respect to the first housing part via the at least one first sensor.

17. The foldable electronic device of claim 16, wherein identifying whether the third housing part is at least partially folded with respect to the first housing part via the at least one first sensor is disabled based on identifying that the first housing part and the second housing part are folded via the at least one second sensor.

18. The foldable electronic device of claim 1, wherein the instructions, when executed by the at least one processor individually or collectively, cause the foldable electronic device to, based on identifying that the third housing part is at least partially folded with respect to the first housing part, enable identifying whether the first housing part and the second housing part are unfolded or folded, and wherein identifying whether the first housing part and the second housing part are unfolded or folded is disabled based on identifying that the third housing part is not at least partially folded with respect to the first housing part.

19. The foldable electronic device of claim 1, wherein the second housing part is, when the first housing part and the second housing part are folded, rotated in a first rotational direction with respect to the first housing part, and wherein the third housing part is, when the first housing part and the third housing part are folded, rotated in a second rotational direction opposite to the first rotational direction with respect to the first housing part.

20. The foldable electronic device of claim 1, wherein a width of the first hinge is different from a width of the second hinge.
